Exploring the conjunction fallacy in probability judgment: Conversational implicature or extension neglect?

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EXPLORING THE CONJUNCTION FALLACY IN PROBABILITY JUDGMENT: CONVERSATIONAL IMPLICATURE OR EXTENSION NEGLECT?*

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According to the conjunction rule of probability theory, a conjunction of events cannot be more probable than either conjunct, i.e. $P(A\&B) \leq P(A), P(B)$. However, participants often violate this rule in experimental settings, thus committing the conjunction fallacy. Why do participants commit this fallacy? One hypothesis suggests that participants interpret a task-critical statement designating the event A as designating the event $A\&notB$. If so, participants do not commit the conjunction fallacy. Another hypothesis suggests that participants fail to take task-relevant relations of set inclusion into account when judging probabilities. Both hypotheses were tested in an experiment utilizing a between-subjects design with 145 participants. The results, analyzed using Bayes factors, provide evidence for the null hypothesis in both cases, thus suggesting that neither hypothesis explains the fallacy. However, the unexpectedly low prevalence of the fallacy in the baseline group may have masked the true effects of the manipulations.

Thinking, construed in a sufficiently broad sense, is the act or process of performing mental operations on the contents of one's mind (Holoyak & Morrison, 2005). One might engage in thinking for a variety of reasons; the most noteworthy, perhaps, being that it allows one to gain cognitive access to new information. For example, if one were to know that John was born in New York City, then learning that Mary was born in the same city as John would allow one to conclude that Mary too was born in New York City. Naturally, the adaptive value of this ability can hardly be understated. In general, however, people in real-life situations are not merely interested in gaining access to new information, but in gaining access to new information that is, in some sense, *correct*, *valid* or *true*. Fortunately, there are many tools available that guarantee that the information one gains access to via thinking meets one or several of these criteria. Such tools commonly come in the form of *rules*. For example, if one wishes to correctly calculate the sum of two numbers, one would employ the rules of arithmetic; if one wishes to correctly derive a logical consequence from a set of premises, one would apply the inference rules prescribed by the suitable logic, and if one wishes to correctly judge the probability of an event occurring, one would employ the rules of probability theory. Such rules can be regarded as constituting the normative standards of thought—they designate, for all intents and purposes, and to the best of our knowledge, the proper and epistemically rational ways of thinking and reasoning. Hence, whenever people aspire to valid reasoning or judgment, they are rational only insofar as their thought processes conform to such normative standards.

For researchers interested in studying thought, reasoning, rationality, or some derivative phenomenon, two fundamental questions emerge: (i) Which rules constitute the normative standards of thought? and (ii) To what capacity do people adhere to those

rules in practice? The first question is mainly of concern to philosophers, logicians and mathematicians; the second question, by contrast, is studied in depth primarily by researchers in cognitive psychology. With regards to the second question, cognitive psychologists have amassed a substantial body of research demonstrating that people, on certain tasks, in fact do systematically fail to think and reason in accordance with normative standards (e.g. Gilovich, Griffin & Kahneman, 2002). Such failures include, among other things, the various fallacies people tend to commit during probability judgment (Bar-Hillel & Neter, 1993; Tversky & Kahneman, 1983), misperceptions of randomness (Bar-Hillel & Wagenaar, 1991), failures to use the correct inference rule when testing hypotheses (Wason, 1960), and invalid attributions of causality (Sloman & Lagnado, 2015). Such systematic failures to adhere to normative standards have raised some doubt among researchers concerning people's capacities for rational thought. While a number of such systematic failures have been studied, it is undoubtedly the case that few have received as much scholarly attention as the conjunction fallacy.

The conjunction fallacy in probability judgment

According to probability theory, it can never be more probable that two events occur together than that one of the events occurs. For instance, if one rolls two fair six-sided dice, it cannot be more probable that one rolls a six on both dice than that one rolls a six on one die (irrespective of what one rolls on the other die). Similarly, it cannot be more probable that John is both a carpenter and a father than it is that John is a father (irrespective of whether he is a carpenter or not). This fact is due to a rule of probability theory generally referred to as 'the conjunction rule' (CR):

(CR) For any two events A and B, it necessarily holds that the probability of the conjunction $A\&B$ cannot exceed the probability of either constituent event, i.e. it necessarily holds that $P(A\&B) \leq P(A)$ and that $P(A\&B) \leq P(B)$.[1]

In other words, (CR) is a normative standard of probability theory: Any agent who aspires to make valid and rational probability judgments needs to make those probability judgments in accordance with (CR), or else the judgments are determinately incorrect. By contrast, if an agent violates (CR) by judging a conjunction of events $A\&B$ to be more probable than at least one of its constituent events, that agent is said to have committed *the conjunction fallacy* (Tversky & Kahneman, 1983).

Strikingly, research shows that a certain class of judgment tasks reliably lead participants to commit the conjunction fallacy (Fantino, Kulik, Stolarz-Fantino & Wright, 2003; Stolarz-Fantino, Fantino, Zizzo & Wen, 2003; Tversky & Kahneman, 1983). As (CR) is generally taken to be one of the most basic and fundamental qualitative rules of probability theory, the fact that participants frequently violate this rule in experimental settings has put people's capacity for rational probability judgment in question. While the judgment tasks used to study the conjunction fallacy come in several variants, the most common format consists of presenting a personality

[1] The validity of (CR) is established by deriving a suitable mathematical regimentation of the rule from the so-called 'Kolmogorov axioms', which serve as the foundation for standard conceptions of probability theory (Alm & Britton, 2008).

description of a fictitious person (henceforth called a 'vignette') together with a list of statements (henceforth called 'event-statements') conditioned on that vignette. After reading the vignette, participants are asked to judge the probabilities of each event-statement, either by ranking them (from most probable to least probable) or by providing numerical estimates. The following is an example of such a judgment task:

> *Linda is 31 years old, single, outspoken and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in anti-nuclear demonstrations. Please rank the following statements in order of their respective probability.*
>
> ____ *Linda is a teacher in elementary school.*
> ____ *Linda works in a bookstore and takes Yoga classes.*
> ____ *Linda is active in the feminist movement.*
> ____ *Linda is a psychiatric social worker.*
> ____ *Linda is a member of the League of Women Voters.*
> ____ *Linda is a bank teller.*
> ____ *Linda is an insurance salesperson.*
> ____ *Linda is a bank teller and is active in the feminist movement.*

This famous judgment task is known as 'the Linda problem' (Tversky & Kahneman, 1983). According to (CR), the conjunction 'Linda is a bank teller and is active in the feminist movement' cannot be more probable than its constituent event-statement 'Linda is a bank teller'. However, in a study by Tversky and Kahneman (1983), 89% of the participants committed the conjunction fallacy by ranking the event-statement 'Linda is a bank teller and is active in the feminist movement' as the more probable of the two. Furthermore, in the same study, statistically sophisticated participants (graduate students in statistics) committed the conjunction fallacy in almost equal proportion (85%) to statistically naïve participants (89%), and various attempts by the two authors to reduce the prevalence of the conjunction fallacy by rephrasing or simplifying the judgment tasks were remarkably unsuccessful. Since Tversky and Kahneman's (1983) original study, many studies have replicated and corroborated their findings (e.g. Fiedler, 1988; Sides, Osherson, Bonini & Viale, 2002; Stolarz-Fantino, Fantino et al., 2003), and it is widely agreed that the conjunction fallacy is a robust phenomenon.

What motivates researchers in studying the conjunction fallacy?

One might wonder why researchers find the conjunction fallacy so interesting. Surely, people commit any number of mathematical and logical errors on a regular basis—what distinguishes the conjunction fallacy from other such errors people might commit? To this question, there are three main answers. First and foremost, the tendency to commit the conjunction fallacy is remarkably robust, and a high prevalence of conjunction fallacies are found in virtually every study investigating the fallacy. This indicates that there is indeed some cognitive mechanism (or some array of mechanisms) that causes participants to systematically make these errors. Secondly, it is often the case in psychological research that the study of various cognitive or perceptual mechanisms is facilitated by discovering instances where those mechanisms produce errant or illusory results. In this sense, researchers studying the conjunction fallacy believe that by gaining a better understanding of the conjunction fallacy, one would thereby gain a

better understanding of the mechanisms through which the mind produces probability judgments. Last but not least, there is the tacit assumption that people's tendency to commit the conjunction fallacy may adversely affect real-life decision making. If the conjunction fallacy is a robust, ecologically valid phenomenon, there might be a wide array of situations in which people make poor decisions as a result of committing conjunction fallacies. For this reason, many researchers hope that gaining a better understanding of the cognitive mechanisms that produce the conjunction fallacy opens up the possibility to develop tools or strategies that allow people to avoid making the fallacy, thus protecting them from poor decision-making (Tversky & Kahneman, 1983).

Why do people commit the conjunction fallacy?

There are currently several competing accounts on offer attempting to explain people's tendency to commit the conjunction fallacy. The most well-known account (and the one most commonly printed in textbooks) is the representativeness-heuristic hypothesis, originally proposed by Tversky and Kahneman (1983). Although the present study does not investigate the representativeness-heuristic hypothesis *per se*, talk of representativeness is ubiquitous in the research on probability judgment, and thus a brief description of the representativeness-heuristic hypothesis is warranted. Representativeness, described in Tversky's and Kahneman's terms, is 'an assessment of the degree of correspondence between a sample and a population, an instance and a category, an act and an actor or, more generally, between between an outcome and a model' (Kahneman & Tversky, 1972; Tversky & Kahneman, 1983). For example, many would presumably consider the description 'birds that can mimic human speech' to be representative of the category of parrots, but unrepresentative of the category of doves. According to Tversky and Kahneman, people are guided by their assessments of representativeness when judging probabilities (Tversky & Kahneman, 1983). For instance, if asked 'Elsa is a bird that can mimic human speech—what type of bird is she?', many would judge it probable that Elsa is a parrot, but improbable that Elsa is a dove, since the description offered is representative of the category of parrots (and unrepresentative of the category of doves). This is an example of the representativeness heuristic at work. However, while the representativeness heuristic is an economical and often highly successful strategy for judging probabilities under uncertainty, Tversky and Kahneman (1983) argue that it suffers from a fundamental flaw in that assessments of representativeness are not constrained by the rules of probability theory. Hence, a conjunction of events can be assessed as being more representative of its target model than either (or both) of its constituent events, thus leading one to assign a higher probability to the conjunction than to one (or both) of its constituents. According to Tversky and Kahneman (1983), this is precisely what happens in judgment tasks such as the Linda problem: The conjunction 'Linda is a bank teller and is active in the feminist movement' is assessed as being more representative of the target model (i.e. the fictitious person Linda) than its constituent event-statement 'Linda is a bank teller', and hence participants tend to assign a higher probability to the conjunction than to the constituent.

As for its current empirical status, it suffices to say that the representativeness-heuristic hypothesis has become increasingly disputed in recent years. While the hypothesis holds a strong appeal in that it provides a plausible explanation not only for the conjunction fallacy but for an array of observed phenomena relating to subjective

probability judgment, such as misperceptions of randomness, the gambler's fallacy, and base-rate neglect (Kahneman & Tversky, 1972; Tversky & Kahneman, 1983), concerns about the validity of the hypothesis were raised when Gavanski and Roskos-Ewoldsen (1991) showed that participants commit the conjunction fallacy even in judgment tasks where assessments of representativeness are blocked or simply not applicable. Hence, it appears that the representativeness-heuristic hypothesis can at best account only for a subset of all occurrences of the conjunction fallacy, and that other hypotheses regarding the causes of the conjunction fallacy must be considered as well.

The conversational-implicature hypothesis

Some researchers have suggested that, in the context of the judgment tasks, participants may be interpreting the event-statement designating the event A as designating the event $A\&notB$—for instance, interpreting the event-statement 'Linda is a bank teller' as suggesting that Linda is a bank teller who is not active in the feminist movement (Dulany & Hilton, 1991; Fiedler, 1988; Hertwig & Gigerenzer, 1999; Morier & Borgida, 1984; Politzer & Noveck, 1991). If true, then participants do not actually commit any conjunction fallacies, for while it may appear to the experimenters as if participants judge the event-statement 'Linda is a bank teller and is active in the feminist movement' as being more probable than the event-statement 'Linda is a bank teller' (i.e. participants judge that $P(A\&B) > P(A)$) , they are in fact judging the event-statement 'Linda is a bank teller and is active in the feminist movement' as being more probable than the event-statement 'Linda is a bank teller, but she is not active in the feminist movement' (i.e. participants judge that $P(A\&B) > P(A\&notB)$). This seemingly inconsequential difference is in fact highly important, as it absolves the participants of committing the conjunction fallacy. This fact is easily recognized when it is pointed out that participants are then not judging a conjunction of events as being more probable than one of its constituent events, but rather judging one conjunction of events to be more probable than another conjunction of events. Such a judgment does not violate (CR), and does not constitute a conjunction fallacy. In sum, according to this hypothesis, the alleged conjunction fallacies observed by researchers are not actual fallacies, but rather simple linguistic misunderstandings between experimenters and participants.

Clearly, this hypothesis challenges much of the orthodoxy concerning the conjunction fallacy. One might, however, ask what justification there is for the belief that participants interpret statements in this way. With regards to this question, researchers supporting the hypothesis typically appeal to the notion of *conversational implicature*. In brief, 'conversational implicature' is a technical term in the field of pragmatics referring to those aspects of the meaning of a sentence that are not conveyed by the literal meaning of the sentence itself, but rather by how the sentence is used in a conversational exchange governed by certain implicit conversational rules (Grice, 1975). For example, if one asks, 'Is Daniel a top student?', and receives the reply 'Daniel certainly tries his best to hand in his assignments on time', one would presumably draw the conclusion that Daniel is not a top student (but perhaps a rather mediocre one). Note, however, that this meaning is merely suggested by the answer, rather than literally stated. Hence, this meaning that is suggested (rather than literally stated) constitutes a conversational implicature. While the research literature on conversational implicature can be technical, it can be loosely described as operating on

the foundational assumption that speakers and hearers engaged in conversation expect each other to obey certain conversational rules, or *maxims*, in order to facilitate effective communication. Table 1 contains the original conversational maxims as suggested by Grice (1975).

Table 1. Conversational maxims as suggested by Grice (1975).

Name of maxim	Description
Maxim of Quality	Try to make your contribution one that you believe to be true, specifically: Do not say what you believe to be false. Do not say that for which you lack adequate evidence.
Maxim of Quantity	Make your contribution as informative as is required for the current purposes of the exchange. Do not make your contribution more informative than is required.
Maxim of Relevance	Make your contributions relevant.
Maxim of Manner	Avoid obscurity. Avoid ambiguity. Be brief. Be orderly.

If hearers indeed assume that speakers obey such maxims, then it is easy to see how the sentence 'Linda is a bank teller' might be interpreted as suggesting that she is a bank teller that is not active in the feminist movement. Assume, for example, that the following two statements are asserted:

(i) Linda is a bank teller.
(ii) Linda is a bank teller who is active in the feminist movement.

A hearer assuming that the maxim of quantity is obeyed may take it that the speaker who asserts (i) actually means to assert that Linda is a bank teller who is not active in the feminist movement, but that the speaker omits the subclause 'who is not active in the feminist movement' in order to avoid making the contribution more informative than is required for the exchange. Thus, the hypothesis receives some *prima facie* justification on theoretical grounds, as it is founded on established theory in the field of pragmatics. While the line of reasoning here presented may certainly be challenged on theoretical grounds by researchers knowledgeable in pragmatics, this study will take the empirical route and evaluate the hypothesis that alleged conjunction fallacies are caused by conversational implicatures—henceforth called 'the conversational-implicature hypothesis'—by testing the hypothesis empirically.

With regards to empirical testing, several authors have attempted to test the conversational-implicature hypothesis by attempting to block the implicature in various creative ways to see if this results in fewer conjunction fallacies. In Tversky and Kahneman (1983), an attempt was made to avoid the implicature by replacing the sentence 'Linda is a bank teller' with the sentence 'Linda is a bank teller whether or not she is active in the feminist movement'. With this variation, the proportion of conjunction fallacies dropped to 59%. Similarly, Messer and Griggs (1993) attempted to block the implicature by changing the sentence 'Linda is a bank teller' to 'Linda is a bank teller regardless of whether or not she is active in the feminist movement' and

observed a drop in the proportion of conjunction fallacies to 56%. Furthermore, Morier and Borgida (1984) tried to block the implicature by having one group of participants judge probabilities on standard judgment tasks while another group judged probabilities on so-called 'debiased judgment tasks'. Debiased judgement tasks were identical to standard judgment tasks, except that a statement corresponding to the event $A\&notB$ had been explicitly added to the list of statements to be judged for probability. For example, here is a debiased version of the Linda problem:

> *Linda is 31 years old, single, outspoken and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in anti-nuclear demonstrations. Please rank the following statements in order of their respective probability.*
>
> ____ *Linda is a bank teller.*
> ____ *Linda is a bank teller and is active in the feminist movement.*
> ____ *Linda is active in the feminist movement.*
> ____ *Linda is a bank teller who is not active in the feminist movement.*

Again, if hearers assume that speakers obey the maxim of quantity, then hearers ought to interpret the sentences 'Linda is a bank teller' and 'Linda is a bank teller who is not active in the feminist movement' as conveying different meanings, for if the speaker would intend the two sentences to be interpreted the same way, then the speaker would be taken as repeating himself for no apparent reason, thus violating the maxim of quantity by making the contribution to the exchange more informative than is required. Hence, it follows from theory on conversational implicature that adding the sentence 'Linda is a bank teller who is not active in the feminist movement' would lead participants to interpret the sentence 'Linda is a bank teller' as intended by experimenters, i.e. as being compatible with both the assertion that Linda is active in the feminist movement and the assertion that she is not active in the feminist movement. Returning to Morier and Borgida (1984), the prevalence of the conjunction fallacy was reduced from 77% on the standard version of a judgment task (the so-called 'Bill problem') to 49% on the debiased version of the same task. However, when judging probabilities on a different debiased judgment task (the Linda problem), the proportion of participants who committed the conjunction fallacy (77%) was not substantially lower when compared to participants who received the standard version of that task (80%). Additionally, in a study by Dulany and Hilton (1991), participants were asked to provide probability judgments on the standard Linda problem, and were later asked whether or not they interpreted the sentence 'Linda is a bank teller' as suggesting that she is a bank teller but not a feminist. While the prevalence of the conjunction fallacy reached 52%—a comparatively low proportion when compared to the results reported by Tversky and Kahneman (1983)—only 25% of participants reported that they did indeed interpret the sentence in this manner.

In sum, previous research indicates that attempts to block the implicature sometimes results in a substantially reduced proportion of conjunction fallacies, and sometimes not. Despite of this, the hypothesis retains a high degree of confidence among some influential researchers (e.g. Hertwig & Gigerenzer, 1999). While the previous research indicates that the hypothesis hardly can explain all instances of the conjunction fallacy,

results are consistent with the assumption that at least a substantial proportion of conjunction fallacies can be explained by appeal to the implicature. It is thus useful to distinguish between a *strong form* of the conversational-implicature hypothesis (which states that virtually all conjunction fallacies can be explained by appeal to the implicature) and a *weak form* of the hypothesis (which states that a substantial proportion, but not all, conjunction fallacies can be explained by appeal to the implicature). The present study thus seeks contribute to the scientific discussion regarding the weak form of the conversational-implicature hypothesis by conducting an experiment in which the results from previous studies are aggregated and used as a basis for constructing a prior (H1) for Bayesian data analysis. By calculating Bayes factors, it is possible to diagnose whether the data collected in this experiment supports H1 or the null hypothesis (H0). If the collected data supports H1, this will be taken as corroborating (the weak form of) the conversational-implicature hypothesis.

The extension-neglect hypothesis

One interesting result in the research literature on the conjunction fallacy is that participants tend to commit fewer conjunction fallacies when the the response format of the judgment tasks is changed to involve frequencies rather than probabilities. For instance, Tversky and Kahneman (1983) presented the following judgment task to a group of participants:

> *A health survey was conducted in a sample of 100 adult males in British Columbia, of all ages and occupations. Please give your best estimate of the following values:*
>
> *(1) How many of the 100 participants have had one or more heart attacks ?*
>
> *(2) How many of the 100 participants both are over 55 years old and have had one or more heart attacks ?*

Only 25% of the participants committed the conjunction fallacy by estimating a lower value for the first alternative than for the second alternative. When presenting the judgment task to a different group of participants, the prevalence of the conjunction fallacy dropped to a record low of 11%. Tversky and Kahneman (1983) thus speculate that changing the response format to involve frequencies (rather than probabilities) aides the participants in correctly representing the relation of set inclusion obtaining between the sets designated by the judgment task. To make this stipulation clear, notice that two task-relevant sets are designated in the judgment task presented above: (1) The set of participants who have had one or more heart attacks, and (2) the set of participants who both are over 55 years old and have had one or more heart attacks. The relation of set inclusion obtaining between these sets can be illustrated with a Venn diagram, as in figure 1. As is clear from figure 1, the set *A*&*B* is included in the set *A*. Once this fact is appreciated, it becomes becomes virtually self-evident that the set *A*&*B* cannot contain more elements than the set *A*. Hence, Tversky and Kahneman (1983) conclude that a correct representation of the task-relevant set inclusions is facilitated when the response format includes a frequency response format rather than a probability response format, and that participants who correctly represent this set inclusion thus avoid committing the conjunction fallacy.

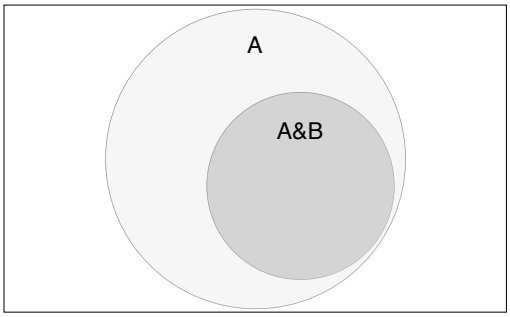

Figure 1. Venn diagram illustrating the relation of set inclusion obtaining between the set of participants who have had one or more heart attacks (*A*) and the set of participants who both have had one or more heart attacks and are over 55 years old (*A*&*B*).

This conclusion is interesting, as it shifts a large part of the blame for the conjunction fallacy from the representativeness-heuristic to the neglect of set inclusions, at least on judgment tasks with a frequency response format. However, Tversky and Kahneman (1983) suggest that the same holds for judgment tasks with probability response formats. Henceforth, the hypothesis that participants who correctly represent the task-relevant set inclusions will avoid committing the conjunction fallacy will be referred to as 'the extension-neglect hypothesis'.

One can test the extension-neglect hypothesis empirically by attempting to evoke correct cognitive representations of the task-relevant set inclusions in order to see if this results in a drastically reduced proportion of conjunction fallacies when compared to a baseline. Specifically, the mean prevalence of the conjunction fallacy on standard judgment tasks with a probability response format is approximately 70%. It will thus be stipulated that the extension-neglect hypothesis predicts the proportion of conjunction fallacies to drop to about 10% (as in Tversky & Kahneman, 1983) when correct representations of set inclusions are in place. The perhaps most simple and direct way to evoke correct representations of set inclusion is to present a Venn diagram with each judgment task illustrating the set inclusions. One would expect of the participants that they, upon seeing the Venn diagrams, would not only correctly represent the task-relevant set inclusions, but also infer that these set relations are relevant to the probability judgments they are asked to perform. As with the conversational-implicature hypothesis, the prediction is represented as a prior (H1), and the data obtained from testing the extension-neglect hypothesis in this experiment are analyzed using Bayes factors. Ideally, this allows one to diagnose whether H0 or H1 is supported by the collected data.

The aims of this study

In sum, the present study tests both the conversational-implicature hypothesis and the extension-neglect hypothesis experimentally by utilizing a between-subjects design involving three groups (a baseline group and two experimental groups). One experimental group tests the conversational-implicature hypothesis, and the other tests the extension-neglect hypothesis. Each hypothesis is tested independently using Bayes factors to assess whether its designated prior (H1) or the null hypothesis (H0) is supported by the collected data.

Method

Participants

A total of 145 participants (mean age = 26.8 years, *s* = 8 years) were recruited via convenience sampling. 48 participants stated that they were male, 95 participants stated that they were female, and 2 participants stated that they consider themselves to have a non-binary gender identity. Out of the 145 participants recruited, 91 were undergraduate psychology students. These participants were recruited in one of two ways: The first method of recruitment involved sending an e-mail to all psychology students enrolled in psychology classes at Stockholm university, inviting them to partake in the experiment in exchange for course credit. The second method of recruitment involved handing out the experimental materials during psychology lectures and seminars, again offering course credit in exchange for participation. The remaining 54 participants were not psychology students. These participants were recruited via the author's personal Facebook profile, where links were posted to online versions of the experimental materials. These participants were offered a lottery ticket, where the lottery prize was a gift certificate worth 500 SEK at a Swedish bookstore, as a reward for participation. All participants were explicitly informed that their participation in the experiment was voluntary, that they were free to end their participation at any time without consequence, and that the methods of data collection and analysis would provide them with complete anonymity. However, no written informed consent was collected.

Once recruited, participants were randomized to one out of three groups using the true random number generator at https://www.random.org/. One group served as a baseline group, whereas the two other groups, here termed ‘the debias group’ (which tested the conversational-implicature hypothesis) and ‘the Venn group’ (which tested the extension-neglect hypothesis) respectively, served as experimental groups. Hence, using this design, it was possible to test both the conversational-implicature hypothesis and the extension-neglect hypothesis in the same experiment.

Materials

The experimental materials for this study were constructed around two judgment tasks, which will be referred to as 'the Linda problem' and 'the Stefan problem' respectively. While the Linda problem used in this study was copied almost verbatim from Tversky and Kahneman (1983), the Stefan problem was a novel problem created solely for this study. Furthermore, three versions of each judgment task were created: A standard version, a debiased version, and a Venn version. All experimental materials were presented in Swedish. The original Swedish versions of all judgment tasks can be found in the appendix.

The standard versions of the judgment tasks each consisted of a vignette followed by three event-statements conditioned on the vignette. Below is the standard version of the Linda problem, translated into English:

> *Linda is 31 years old, single, outspoken, and very bright. At university she majored in gender studies. As a student, she engaged herself in questions concerning discrimination and social justice, and she participated in demonstrations against racism.*

Please judge the probabilities of the statements written below. Probabilities are to be written as percentages (where 0% is the lower bound, and 100% is the higher bound).

____ *Linda is a bank teller.*
____ *Linda is a bank teller and a feminist.*
____ *Linda is a feminist.*

Additionally, below is the standard version of the Stefan problem, again translated into English:

Stefan is 54 years old. He is intelligent, but very quiet. He cultivated a large interest in mathematics and computers already as a child, and today he holds an engineering degree with a major in computer technology.

Please judge the probabilities of the statements written below. Probabilities are to be written as percentages (where 0% is the lower bound, and 100% is the higher bound).

____ *Stefan is a schlager artist.*
____ *Stefan is a schlager artist and a programmer.*
____ *Stefan is a programmer.*

As for the the debias versions of the judgment tasks, these were identical to the standard versions, with the exception that an event-statement corresponding to $A\&notB$ was added to the list of statements to be judged for probability. Below is the debias version of the Linda problem:

Linda is 31 years old, single, outspoken, and very bright. At university she majored in gender studies. As a student, she engaged herself in questions concerning discrimination and social justice, and she participated in demonstrations against racism.

Please judge the probabilities of the statements written below. Probabilities are to be written as percentages (where 0% is the lower bound, and 100% is the higher bound).

____ *Linda is a bank teller.*
____ *Linda is a bank teller and a feminist.*
____ *Linda is a feminist.*
____ *Linda is a bank teller, but not a feminist.*

The purpose of including the event-statement 'Linda is a bank teller, but not a feminist' was to block the assumed conversational implicature leading participants to interpret the event-statement 'Linda is a bank teller' as meaning that Linda is a bank teller but not a feminist. The method of blocking the implicature here employed is thus the same as in

Morier and Borgida (1984). The alteration made to the Stefan problem was equivalent to that made to the Linda problem.

Finally, the Venn versions of the judgment tasks were also identical to the standard versions, except that each judgment task was presented together with a Venn diagram emphasizing the task-relevant set inclusions. For the Linda problem, this amounted to emphasizing the set inclusion obtaining between the set of bank tellers and the set of feminist bank tellers. For the Stefan problem, by contrast, it amounted to emphasizing the set inclusion obtaining between the set of programmers and the set of programmers. The Venn diagram presented together with the Linda problem is depicted in figure 2.

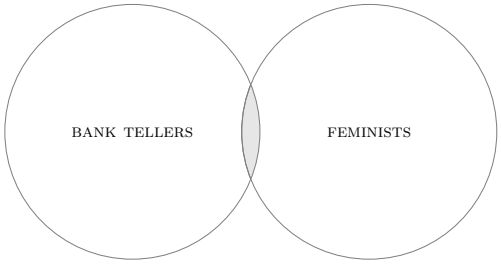

Figure 2. Venn diagram emphasizing the set inclusion obtaining between the set of bank tellers and the set of feminist bank tellers.

Given these judgment tasks, three different surveys were created. The standard survey contained the standard versions of two judgment tasks, whereas the debias survey contained the debiased versions of the judgment tasks, and the Venn survey contained the Venn versions of the judgment tasks. Table 2 provides an overview of the structure of the experiment.

Table 2. Overview of the structure of the experiment.

Group	Survey	Predictions according to the hypotheses
Baseline group	Standard survey	Participants will commit the conjunction fallacy.
Debias group	Debias survey	*Conversational-implicature hypothesis:* Prevalence of the conjunction fallacy will be substantially lower than in the baseline group.
Venn group	Venn survey	*Extension-neglect hypothesis:* Prevalence of the conjunction fallacy will be drastically lower than in the baseline group.

Procedure

For the participants who were recruited via Facebook, the following applies: Online versions of the three surveys were created via the website https://www.surveymonkey.com/. Furthermore, a registry form was created using Google Forms. A link to the registry form was posted on the author's personal Facebook page, together with a brief and purposefully vague description of the study and its purpose, as well as a description of the reward for participation. Each participant

who registered for participation using the registry form was then randomized to one and only one of the three groups using the random number generator at https://www.random.org/. This was performed manually. The participants who were randomized to the baseline group were sent a link via e-mail to the online version of the standard survey, whereas the participants who were randomized to the debias group were sent a link to the online version of the debias survey, and equivalently for the participants randomized to the Venn survey.

With regards to the participants who were undergraduate psychology students: Data was collected in conjunction with psychology lectures and seminars on three different occasions. On each of these occasions, an equal number of standard, debias and Venn surveys were printed in advance, and then put in a pile which was shuffled. Each participant then took a survey from the shuffled pile.

Analysis

The data was analyzed using Bayes factors. The justification for using Bayesian statistics was that a statistical analysis capable of diagnosing support for the null hypotheses (i.e. no difference between groups) was required. The orthodox Neyman-Pearson approach to null hypothesis significance testing (NHST) can never provide support for the null hypothesis, as a non-significant result in NHST can mean either that the null hypothesis is true or that the null hypothesis is false but that the statistical test lacked sufficient power to provide a significant result (e.g. Dienes, 2008). By contrast, Bayesian statistics provides the tools to properly diagnose which hypothesis (H0 or H1) is supported by the data. Hence, the use of Bayesian statistics, rather than NHST, is well-motivated in this scenario.

In this study, the Bayesian analysis was conducted using natural logarithms of odds ratios as suggested by Beard, Dienes, Muirhead and West (2016). By taking the natural logarithms of odds ratios, those odds ratios are transformed into an approximate normal distribution, thus allowing for the creation of approximately normally distributed priors. Priors were constructed as follows:

With regards to the conversational-implicature hypothesis: According to the null hypothesis (H0), there should be no difference in the prevalence of the conjunction fallacy between the baseline group and the debias group, i.e. the odds ratio (OR) between the groups should be OR = 1. Hence the null hypothesis predicts a point estimate equal to the natural logarithm of OR = 1, which is 0. For the experimental hypothesis (H1), results from previous studies testing this hypothesis were considered (Messer & Griggs, 1993; Morier & Borgida, 1984; Tversky & Kahneman, 1983). Aggregating the relevant results from those studies, the mean OR between the baseline groups (receiving standard judgment tasks) and experimental groups (receiving debiased judgment tasks) adds up to 2.81, the natural logarithm of which is 1.03. Thus, it was decided that H1 should be constructed using a half-normal distribution centered on 0 (the natural logarithm of OR = 1) with a standard deviation of 1.03. Essentially, this prior expresses that, in terms of odds ratios, the true effect of blocking the implicature is likely somewhere between 1 and 2.81. Additionally, as H1 is constructed using a half-normal distribution centered on 0, a negative result (a negative log OR) is deemed impossible.

With regards to the extension-neglect hypothesis: Again, the null hypothesis (H0) predicts a point estimate of 0 (the natural logarithm of OR = 1). As for H1, it was first considered that the mean prevalence of the conjunction fallacy on standard tasks as reported in previous studies is approximately 70% (i.e. the mean odds of committing the conjunction fallacy are approximately 2.33). If, as the extension-neglect suggests, a failure to correctly represent task-relevant set inclusions is the primary cause of the conjunction fallacy, then a plausible prediction would be that the prevalence of the conjunction fallacy would drop to about 10% (i.e. the odds of committing the fallacy would drop to 0.11) in the Venn group. The OR thus amounts to 2.33/0.11 = 21, the natural logarithm of which is 3.04. An estimated population standard deviation would be 3.04/2 = 1.52. Thus, the prior for the extension-neglect hypothesis assumes a log-normal distribution centered on 3.04 with a standard deviation of 1.52. Essentially, this prior expresses that the population mean effect is OR = 21, i.e. that the odds of committing the fallacy when representations of set-relations are evoked are on average 21 times greater than when they are not evoked.

Results

Out of the total 145 participants, 138 participants completed their participation in full by providing probability judgments on both judgment tasks. All data pertaining to the remaining 7 participants who did not complete their participation in full were removed from the data analysis.

With regards to the prevalence (i.e. relative frequency) of the conjunction fallacy, the judgments provided on the Linda and Stefan problems in each group were pooled. Results are presented in the table 3.

Table 3. Number of observed conjunction fallacies in each group, and relative frequencies. Note that each participant completed two judgment tasks; hence the relative frequency in each group was calculated by dividing the number of observed fallacies in the group by the total number of judgment tasks completed in that group.

	Conjunction fallacies	
Group	Frequency	Relative frequency (%)
Baseline group (n = 53)	41	39%
Debias group (n = 35)	30	43%
Venn group (n = 50)	30	30%

As a measure of effect size with regards to the effectiveness of the manipulation (i.e. presenting debias problems instead of standard problems, or presenting Venn problems instead of standard problems), odds ratios (OR:s) were calculated. When comparing the baseline and debias groups, the OR was 0.84, indicating that the odds for committing the conjunction fallacy in the baseline group were 0.84 times greater in the baseline group than in the debias group. When comparing the Venn and baseline groups, the OR was 1.47, indicating that the odds for committing the conjunction fallacy were 1.47 times greater in the baseline group than in the Venn group.

In order to perform the Bayesian data analysis, the odds ratios were first transformed by taking their natural logarithms. Bayes factors were then calculated and interpreted in terms of the verbal categories suggested by Jeffreys (1998). With regards to the debias group (as compared to the baseline group), the Bayes factor was $BF_{10} = 0.2$, indicating substantial evidence for the null hypothesis. With regards to the Venn group (as compared to the baseline group), the Bayes factor was $BF_{10} = 0.1$, indicating again substantial evidence for the null hypothesis.

Discussion

It is hardly controversial to state that the conjunction fallacy is an elusive phenomenon. Despite many attempts throughout the years to provide a satisfactory account of the factors that cause conjunction fallacies, no fully satisfactory account has been given. In an attempt to contribute to the discussion on the conjunction fallacy, the present study has evaluated two hypotheses:

(1) *The conversational-implicature hypothesis,* which states that a substantial proportion of observed conjunction fallacies can be explained by the fact that participants interpret the event-statement designating the event A as designating the event $A\&notB$.

(2) *The extension-neglect hypothesis,* which states that almost all observed conjunction fallacies can be explained by participants failing to correctly represent task-relevant relations of set inclusion when judging probabilities.

The results of the present study favor the null hypothesis in both cases.

Prevalence of the conjunction fallacy in the baseline group

It may be noted at the outset that the prevalence of the conjunction fallacy in the baseline group was lower than expected—while the prevalence of the conjunction fallacy on standard tasks is typically around 70-80% (Fantino et al., 1997; Sides et al., 2007; Tversky & Kahneman, 1983), only 39% of the judgments provided by participants constituted conjunction fallacies. Certainly, this number is still high; however, it is substantially lower than the numbers usually found on similar judgment tasks in previous studies. What explains this discrepancy? It is deemed implausible that the discrepancy relates to the verbal formulations of the judgment tasks, as the Linda problem (which typically yields a high proportion of conjunction fallacies) was copied almost verbatim from Tversky and Kahneman (1983), and nothing in the research literature indicates that the Stefan problem, while novel and previously untested, would be any less effective at producing conjunction fallacies. In fact, one of the core results with regards to the conjunction fallacy has been that the fallacy is surprisingly robust with regards to alterations to the formulations of the judgment tasks, even when the alterations have been explicitly tailored in attempt to reduce the proportion of conjunction fallacies (Tversky & Kahneman, 1983). Another explanation deemed equally implausible is that the low prevalence of the conjunction fallacy was due to the statistical sophistication of the participants, which might have allowed them to avoid the

committing the conjunction fallacy. Again, a core result with regards to the conjunction fallacy is that it's prevalence remains high even among highly statistically sophisticated participants, such as graduate students in statistics, who have done a fair degree of coursework in probability theory (Tversky & Kahneman, 1983). Hence, the discrepancy remains unaccounted for.

The conversational-implicature hypothesis

In the present study, the prevalence of the conjunction fallacy in the baseline group was compared to that of the debias group, and the resulting Bayes factor ($BF_{10} = 0.2$) indicated substantial evidence for the null hypothesis. In fact, not only did the data support the null hypothesis, but the prevalence of the conjunction fallacy was actually higher in the debias group (43%) than in the baseline group (39%). Hence, the results in the present study clearly speak against the conversational-implicature hypothesis.

While the results are clear enough, they also present a puzzle: How should these results be interpreted, given that blocking the implicature has been shown to reduce the proportion of conjunction fallacies in previous studies (e.g. Tversky & Kahneman, 1983)? What might account for these differing results? One possible explanation is that blocking the implicature in fact does substantially reduce the prevalence of the conjunction fallacy, but that this effect was masked in the present study due to the unexpectedly low proportion of conjunction fallacies in the baseline group. It is, however, difficult to assess the validity of this explanation. A different explanation was suggested in Morier and Borgida (1984), as they faced a similar problem: It was observed in their study that a debiased version of the Linda problem did not result in a substantially lowered prevalence of the conjunction fallacy (77% vs. 80%), but a debiased version of a different problem ('the Bill problem') did result in a substantially lowered prevalence (49% vs. 77%). This discrepancy needed to be accounted for. They thus speculated (1) that the Linda problem was much more likely to implicate representativeness thinking, and (2) that conjunction problems that strongly implicate representativeness thinking are more resistant to debiasing efforts. If this explanation is correct, one would expect that debiasing the Linda problem would rarely if ever result in a substantially lowered proportion of conjunction fallacies; furthermore, if the Stefan problem, like the Linda problem, strongly implicates representativeness thinking, then this would explain why the results supported the null hypothesis in the present study. However, acceptance of this explanation requires several leaps of faith. First of all, the notion that Linda problem implicates representativeness thinking to a greater degree than, for instance, the Bill problem in Morier and Borgida (1984), is an untested assumption. Secondly, no justification is provided for the assumption that problems which strongly implicate representativeness thinking are more resistant to debiasing. Thirdly, Tversky and Kahneman (1983) and Messer and Griggs (1993) also used a debiased version of the Linda problem—albeit with a different strategy to block the implicature—which did result in substantially fewer conjunction fallacies (59% vs. 85% and 56% vs. 77% respectively). One possible interpretation of the aforementioned results, that is consistent with the results of the present study, is thus that the strategy of blocking the implicature by the inclusion of an event-statement corresponding to $A\&notB$ simply does not reliably block the conversational implicature, and that the lower prevalence of the conjunction fallacy for the Bill problem in Morier and Borgida (1984) can be mainly attributed to other factors, such as random error (for reference,

only n = 30 participants were handed the debiased Bill problem in their study). The fact that Tversky and Kahneman (1983) and Messer and Griggs (1993) saw substantial reductions in the prevalence of the conjunction fallacy may then be attributed to the fact that their strategy of blocking the conversational implicature was, in fact, more robust.

Hence, while the results of the present study supported the null hypothesis, this does not necessarily suggest that the conversational-implicature hypothesis is false. It could be the case that the conversational-implicature hypothesis holds some merit, but that a true effect was masked by the unexpectedly low prevalence of conjunction fallacies in the baseline group, or that the method of blocking the implicature was not reliable. A further rigorous treatment of the conversational-implicature hypothesis would thus presumably involve either (1) empirically justifying the assumption that the inclusion of an event-statement corresponding to $A\&notB$ blocks the implicature before attempting this strategy again, or (2) employing a different strategy which is known to reliably block the implicature. One might also consider testing the conversational-implicature hypothesis using self-report measures, in which participants are asked to report whether or not they interpreted the event-statement designating A as designating $A\&notB$. Indeed, this might be the simplest and most direct way of testing the hypothesis.

The extension-neglect hypothesis

In order to test the extension-neglect hypothesis, the prevalence of the conjunction fallacy in the baseline group was compared to that of the Venn group. The resulting Bayes factor ($BF_{10} = 0.1$) indicates substantial evidence for the null hypothesis.

When interpreting these results, one must keep several things in mind. First of all, while the notion that the Venn diagrams help participants to represent task-relevant relations of set inclusion certainly seems plausible, it might very well be the case that participants fail to correctly understand the diagrams. Secondly, one must again consider the possibility that a true effect of inciting correct representations of task-relevant set inclusions was masked by the unexpectedly low prevalence of the conjunction fallacy in the baseline group. Again, however, it is difficult to judge the validity of this explanation. Thirdly, it should be noted that the experimental hypothesis was very strong in this particular case—the prior estimated an approximately normal distributed centered on $OR = 21$, thus effectively suggesting that if the extension-neglect hypothesis is true, then virtually no one would commit the conjunction fallacy in the Venn group. In this sense, the results, if taken at face value, indicate simply that if the extension-neglect hypothesis is given the momentous task of accounting for virtually all conjunction fallacies, then the data in this experiment serves as substantial evidence that it fails at that task. While this is an interesting result, it does by no means entail that no conjunction fallacies are caused by a failure to represent the task-relevant set inclusions. Indeed, the prevalence of the conjunction fallacy in the Venn group (30%) was in fact noticeably lower than the prevalence in the baseline group (39%), suggesting that it might be the case that emphasizing set inclusions does somewhat reduce the prevalence of the fallacy. However, further experiments would have to be conducted to establish the true size and robustness of this effect.

On a different note, it is important to distinguish between two different variants of the extension-neglect hypothesis: Variant (a), which suggests that participants fail to

correctly represent the task-relevant set inclusions (thus leading them to commit the conjunction fallacy), and variant (b), which suggests that participants neglect to take set inclusions into account (whether they represent them correctly or not). While inciting correct representations of set inclusion by formulating the judgment tasks in terms of frequencies instead of probabilities have been shown to dramatically reduce the fallacy (as in Tversky & Kahneman, 1983), it might be that the cognitive mechanisms that produce intuitive judgments of frequency take set inclusion into account if those set inclusions are recognized, but that the cognitive mechanisms that produce intuitive judgments of probability do not. If variant (b) is the case, and the conjunction fallacy is caused by negligence rather than failure of adequate representation, then evoking correct representations of set inclusion would not be expected to substantially reduce the prevalence of the fallacy.

In sum, if the extension-neglect hypothesis is given the role of explaining the vast majority of conjunction fallacies, then the present study provides evidence against this hypothesis. However, as with the conversational-implicature hypothesis, it might still be that the extension-neglect hypothesis can account for *some* conjunction fallacies. Further experiments are needed to assess whether or not this is the case.

Future research

With regards to the conversational-implicature hypothesis, researchers should establish a robust method that blocks the assumed conversational implicature and continue investigations using that method. Alternatively, self-report measures can be attempted. With regards to the extension-neglect hypothesis, a fair assessment of the role of representations of set inclusion in producing the conjunction fallacy simply mandates more data and more experiments.

As a more general avenue of research, one might consider the fact that while no established explanation of the conjunction fallacy has been found adequate in explaining all observed conjunction fallacies, each proposed explanation has received some degree of empirical justification. Such empirical justification generally consists of data showing that the prevalence of the conjunction fallacy drops whenever the standard judgment tasks are manipulated in some specific way. However, no such manipulation has been found to make the conjunction fallacy disappear completely. It is striking, therefore, that no attempt has been made by researchers to integrate these different explanations into a unified model. Certainly, it could be the case that, in the same sample, some conjunction fallacies can be attributed to, for instance, the representativeness heuristic, while other conjunction fallacies can be attributed to conversational implicature, and yet others to the use of inappropriate rules when combining the probabilities—and so on. It seems reasonably clear that a multi-factor model of the conjunction fallacy would account for a larger proportion of conjunction fallacies than any of the established explanations can account for individually.

Conclusions

With regards to the conversational-implicature hypothesis, the results of the present study supported the null hypothesis. However, it is unclear whether this should be taken as evidence against the conversational-implicature hypothesis, as the true effect might have been masked by the low prevalence of the fallacy in the baseline group.

Furthermore, one cannot rule out that the strategy for blocking the implicature was unreliable. With regards to the extension-neglect hypothesis, the results again favor the null hypothesis, indicating that the conjunction fallacy persists even when task-relevant relations of set inclusion are emphasized. However, it is possible that emphasizing such set inclusions does have a small effect in reducing the prevalence of the fallacy.

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Appendix

Standard judgment tasks

Linda är 31 år gammal, singel, frispråkig, och väldigt skärpt. På universitetet läste hon genusvetenskap. Som student var hon väldigt engagerad i frågor om diskriminering och social rättvisa, och hon deltog aktivt i antirasistiska demonstrationer. Bedöm sannolikheten för nedanstående påståenden. Sannolikheter anges i procent (lägst 0%, högst 100%).

____ Linda är banktjänsteman.

____ Linda är banktjänsteman och feminist.

____ Linda är feminist.

Stefan är 54 år gammal. Han är intelligent, men väldigt tystlåten. Han hade redan som barn ett stort intresse för matematik och datorer, och han har idag en civilingenjörsexamen med inriktning på datateknik. Bedöm sannolikheten för nedanstående påståenden. Sannolikheter anges i procent (lägst 0%, högst 100%).

____ Stefan är schlagerartist.

____ Stefan är schlagerartist och programmerare.

____ Stefan är programmerare.

Debias judgment tasks
Linda är 31 år gammal, singel, frispråkig, och väldigt skärpt. På universitetet läste hon genusvetenskap. Som student var hon väldigt engagerad i frågor om diskriminering och social rättvisa, och hon deltog aktivt i antirasistiska demonstrationer. Bedöm sannolikheten för nedanstående påståenden. Sannolikheter anges i procent (lägst 0%, högst 100%).

____ Linda är banktjänsteman.

____ Linda är banktjänsteman och feminist.

____ Linda är feminist.

____ Linda är banktjänsteman, men inte feminist.

Stefan är 54 år gammal. Han är intelligent, men väldigt tystlåten. Han hade redan som barn ett stort intresse för matematik och datorer, och han har idag en civilingenjörsexamen med inriktning på datateknik. Bedöm sannolikheten för nedanstående påståenden. Sannolikheter anges i procent (lägst 0%, högst 100%).

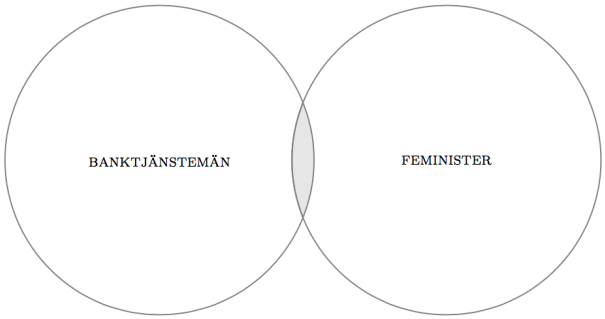
____ Stefan är schlagerartist.

____ Stefan är schlagerartist och programmerare.

____ Stefan är programmerare.

____ Stefan är schlagerartist, men inte programmerare.

Venn judgment tasks



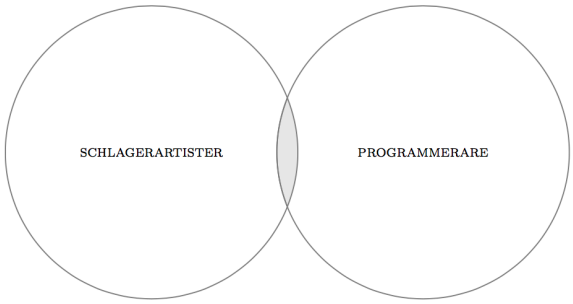
Linda är 31 år gammal, singel, frispråkig, och väldigt skärpt. På universitetet läste hon genusvetenskap. Som student var hon väldigt engagerad i frågor om diskriminering och social rättvisa, och hon deltog aktivt i antirasistiska demonstrationer. Bedöm

sannolikheten för nedanstående påståenden. Sannolikheter anges i procent (lägst 0%, högst 100%).

____ Linda är banktjänsteman.

____ Linda är banktjänsteman och feminist.

____ Linda är feminist.



Stefan är 54 år gammal. Han är intelligent, men väldigt tystlåten. Han hade redan som barn ett stort intresse för matematik och datorer, och han har idag en civilingenjörsexamen med inriktning på datateknik. Bedöm sannolikheten för nedanstående påståenden. Sannolikheter anges i procent (lägst 0%, högst 100%).

____ Stefan är schlagerartist.

____ Stefan är schlagerartist och programmerare.

____ Stefan är programmerare.